

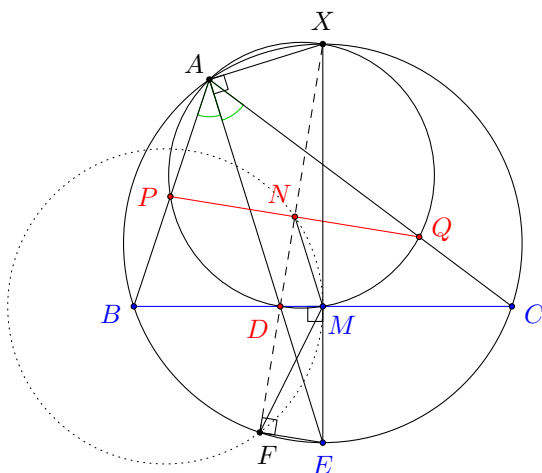
Problem 13 (Poland MO Finals 2015 Problem 6). Let n be a positive integer. Prove that there exists a positive integer $m > n$ such that $1 + 2^m + 3^m$ is divisible by $1 + 2^n + 3^n$.

Solution. We claim that $m = n + \varphi(1 + 2^n + 3^n)$ satisfies the condition. Let $1 + 2^n + 3^n = 2^a 3^b c$ where $(c, 6) = 1$. It suffices to prove $1 + 2^m + 3^m$ is divisible by each of 2^a , 3^b and c .

- If $n = 1$ or n is even, we have $1 + 2^n + 3^n \equiv 2 \pmod{4}$. So $a = 1$. Then it is obvious that $2 \mid 1 + 2^m + 3^m$.
If $n \geq 3$ is odd, we have $1 + 2^n + 3^n \equiv 4 \pmod{8}$. So $a = 2$. Note that $2 = \varphi(4) \mid \varphi(1 + 2^n + 3^n)$. Therefore, m is odd and larger than 1, and thus $4 \mid 1 + 2^m + 3^m$.
Therefore, $2^a \mid 1 + 2^m + 3^m$.
- Note that $b \leq n$, since otherwise $3^n \mid 2^a 3^b c - 3^n = 1 + 2^n$, which is impossible as $3^n > 1 + 2^n$ for all $n > 1$ by simple induction (and when $n = 1$, it is easy to check $b = 1$). Thus, we must have $3^b \mid 3^n$. Also, since $\varphi(3^b) \mid \varphi(1 + 2^n + 3^n) = m - n$, by the Euler-Fermat theorem, we have $1 + 2^m \equiv 1 + 2^n \equiv 0 \pmod{3^b}$. This implies $3^b \mid 1 + 2^m + 3^m$.
- By the Euler-Fermat theorem, we have $1 + 2^m + 3^m \equiv 1 + 2^n + 3^n \equiv 0 \pmod{c}$ since $\varphi(c) \mid \varphi(1 + 2^n + 3^n) = m - n$.

Summarizing all these, we obtain $1 + 2^n + 3^n = 2^a 3^b c \mid 1 + 2^m + 3^m$. □

Problem 14 (USA TSTST 2012 Problem 7). The internal angle bisector of $\angle A$ of $\triangle ABC$ intersects the side BC at D and the circumcircle of $\triangle ABC$ again at E . Let M be the midpoint of BC . The circumcircle of $\triangle ADM$ intersects the sides AB and AC again at P and Q respectively. Let N be the midpoint of PQ . Let F be the foot of perpendicular from E to the line DN . Prove that the line ME is tangent to the circumcircle of $\triangle FMN$.



Solution. (Note that F lies on (ABC) , though we will not prove or use this fact in this solution.)

Let EM meet (ABC) again at X . Note that EX is a diameter of (ABC) , so $\angle EAX = 90^\circ$. As $\angle DAX = \angle DMX = 90^\circ$, the point X also lies on (ADM) . Thus, X is the centre of spiral similarity mapping PQ to BC . As N and M are the midpoints of PQ and BC , the spiral similarity also maps N to M . In addition, since D is the midpoint of arc PQ (as $\angle DAP = \angle DAQ$), while E is the midpoint of arc BC , the spiral similarity maps D to E . Therefore, X, N, D are collinear since X, M, E are collinear.

Now, since $\angle XFE = \angle XMD = 90^\circ$, we have $\triangle XFE \sim \triangle XMD$, and hence $\frac{XF}{XM} = \frac{XE}{XD}$. This is equal to $\frac{XM}{XN}$ by the spiral similarity. It follows that $XM^2 = XN \times XF$, and hence XM is tangent to (MNF) as desired. □

Problem 15 (ARMO 2018 Grade 9 Problem 4). Consider an $n \times n$ chessboard with $n \geq 3$. There is a knight in the top left corner cell. In each step, the knight can move to a new cell such that

- the horizontal distance between the new cell and the old cell is 2 (where the distance of two adjacent cells is 1) and the vertical distance is 1; or
- the vertical distance between the new cell and the old cell is 2 and the horizontal distance is 1.

Three cells A, B, C form a *diagonal triple* if

- the bottom right corner of A coincides with the top left corner of B , and the bottom right corner of B coincides with the top left corner of C ; or
- the top right corner of A coincides with the bottom left corner of B , and the top right corner of B coincides with the bottom left corner of C .

Find all $n \geq 3$ such that we can choose some cells of the chessboard satisfying the following two conditions:

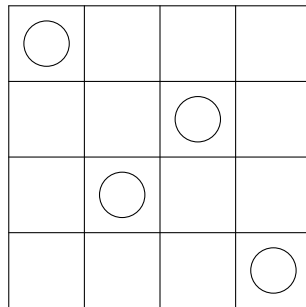
- it is not possible to move the knight to the bottom right corner cell without visiting any chosen cells; and
- for each diagonal triple of the chessboard, at most one cell is chosen.

Solution. n can be any integer greater than 2 which is not congruent to 1 modulo 3.

In the following, we use (i, j) to mean the cell in the i th row (from top to bottom) and the j th column (from left to right). When $n \equiv 0 \pmod{3}$, we choose all cells (i, j) with $j \equiv 2 \pmod{3}$. It is clear that at most one cell is chosen for each diagonal triple. We say that a cell (i, j) is good if i is odd and $j \equiv 0, 1 \pmod{6}$, or i is even and $j \equiv 3, 4 \pmod{6}$. Note that starting from a good cell, one can only move to a good cell if the chosen cells are avoided. (This can be checked manually as there are only a few possible moves.) Since $(1, 1)$ is good but (n, n) is not, one cannot go from $(1, 1)$ to (n, n) .

Similarly, when $n \equiv 2 \pmod{3}$, we choose all cells (i, j) with $j \equiv 0 \pmod{3}$. We say that a cell (i, j) is good if i is odd and $j \equiv 1, 2 \pmod{6}$, or i is even and $j \equiv 4, 5 \pmod{6}$. Then starting from a good cell, one can only move to a good cell. Since $(1, 1)$ is good but (n, n) is not, one cannot go from $(1, 1)$ to (n, n) .

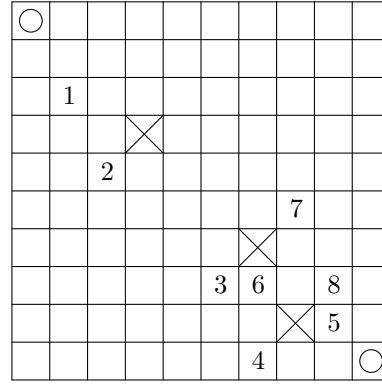
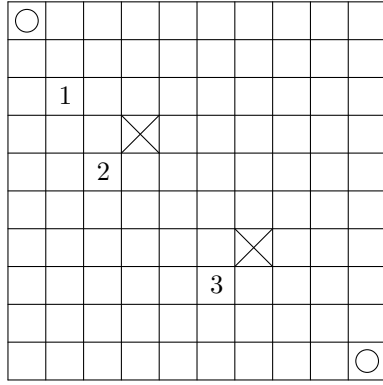
When $n \equiv 1 \pmod{3}$, we prove by induction on n that we can always go from $(1, 1)$ to (n, n) (or in general, from any (i, j) to $(i + n - 1, j + n - 1)$). For the base case $n = 4$, one of $(2, 3)$ and $(3, 2)$ cannot be chosen. So we can always go from $(1, 1)$ to $(4, 4)$ via one of these cells.



Assume we can always go from $(1, 1)$ to (m, m) for any $m \equiv 1 \pmod{3}$ and $4 \leq m < n$. Consider the case n . If $(4, 4)$ is not chosen, we can go from $(1, 1)$ to $(4, 4)$ and then to (n, n) by the inductive hypothesis. Similarly, if $(n - 3, n - 3)$ is not chosen, we can go from $(1, 1)$ to $(n - 3, n - 3)$ and then to (n, n) . Now, assume both $(4, 4)$ and $(n - 3, n - 3)$ are chosen. This implies $(5, 3), (3, 5), (n - 2, n - 4), (n - 4, n - 2)$ are not chosen.

Note that one of $(2, 3)$ and $(3, 2)$ is not chosen. WLOG assume $(3, 2)$ is not chosen. Then we can follow the path $(1, 1) \rightarrow (3, 2) \rightarrow (5, 3) \rightarrow (n - 2, n - 4)$. (The last step corresponds to the inductive hypothesis on the case $n - 6$, or it is the trivial step when $n = 7$.)

Next, if $(n - 1, n - 2)$ is not chosen, we can simply follow the path $(n - 2, n - 4) \rightarrow (n - 1, n - 2) \rightarrow (n, n)$. Now, assume $(n - 1, n - 2)$ is chosen. Then we can follow the path $(n - 2, n - 4) \rightarrow (n, n - 3) \rightarrow (n - 1, n - 1) \rightarrow (n - 2, n - 3) \rightarrow (n - 4, n - 2) \rightarrow (n - 2, n - 1) \rightarrow (n, n)$. It is clear that all these cells cannot be chosen. So we are done.



By induction, all cases $n \equiv 1 \pmod{3}$ with $n \geq 4$ are not solutions. □

Problem 16 (International Zhautykov Olympiad 2019 Problem 2). Find the largest real constant M such that

$$\frac{x_1}{|x_2 - x_3|} + \frac{x_2}{|x_3 - x_4|} + \cdots + \frac{x_{2019}}{|x_{2020} - x_{2021}|} + \frac{x_{2020}}{|x_{2021} - x_1|} + \frac{x_{2021}}{|x_1 - x_2|} > M \quad (1)$$

for any pairwise distinct positive real numbers $x_1, x_2, \dots, x_{2021}$.

Solution. The largest M is 1011.

Firstly, consider

$$x_n = \begin{cases} n\varepsilon^2 & \text{if } n \text{ is odd and } n \neq 2021, \\ 1 - n\varepsilon & \text{if } n \text{ is even or } n = 2021 \end{cases}$$

where $\varepsilon > 0$ is sufficiently small. Clearly, all numbers are distinct. When $\varepsilon \rightarrow 0^+$, we have

$$\begin{aligned} \frac{x_n}{|x_{n+1} - x_{n+2}|} &\rightarrow \frac{0}{1 - 0} = 0 \quad \text{for } n = 1, 3, 5, \dots, 2017, \\ \frac{x_n}{|x_{n+1} - x_{n+2}|} &\rightarrow \frac{1}{1 - 0} = 1 \quad \text{for } k = 2, 4, 6, \dots, 2020, 2021, \\ \frac{x_{2019}}{|x_{2020} - x_{2021}|} &= 2019\varepsilon \rightarrow 0. \end{aligned}$$

Here, $x_{2022} = x_1$ and $x_{2023} = x_2$. It follows that the left-hand side of (1) tends to 1011, so we must have $M \leq 1011$.

Secondly, we prove that (1) holds when $M = 1011$. WLOG assume x_1 is the largest term. We define the indices $i_0 = 1, i_1, i_2, \dots$ such that $x_{i_{k+1}} = \max\{x_{i_k+1}, x_{i_k+2}\}$. Note that i_{k+1} is $i_k + 1$ or $i_k + 2$ (if $x_{i_k+1} = x_{i_k+2}$, we define $i_{k+1} = i_k + 1$). Since x_1 is the largest term, it is clear that there must be a smallest index $t > 0$ such that $i_t = 1$. Then $i_0 < i_1 < i_2 < \cdots < i_{t-1}$, and so the left-hand side of (1) is at least

$$\sum_{j=0}^{t-1} \frac{x_{i_j}}{|x_{i_{j+1}} - x_{i_{j+2}}|} > \sum_{j=0}^{t-1} \frac{x_{i_j}}{x_{i_{j+1}}} \geq t \sqrt[t]{\prod_{j=0}^{t-1} \frac{x_{i_j}}{x_{i_{j+1}}}} = t$$

by the AM-GM inequality. But since each $i_{j+1} - i_j$ (for $j = 0, 1, \dots, t-2$) is at most 2, $i_0 = 1$ and $i_{t-1} \geq 2020$, we must have $t \geq \left\lceil \frac{2020-1}{2} \right\rceil + 1 = 1011$. This proves the bound, and the proof is complete. □